Module selected:

Authentication and Core Employee Module

User stories:

User Story 1 – User Login
As a registered user
I want to log into the system using my credentials
So that I can access system features based on my role

Sub User Stories:

1. Validate user credentials.
2. Redirect user to role-based dashboard.
3. Display error message for invalid login.
4. Maintain secure user session.

User Story 2 – Role-Based Access Control
As an administrator
I want to restrict system features based on user roles
So that unauthorized users cannot access sensitive data

Sub User Stories:

1. Define system roles (Admin, HR, Manager, Employee).
2. Assign permissions to each role.
3. Restrict direct URL access.
4. Display features dynamically according to role.

User Story 3 – Add New Employee
As an HR manager
I want to register a new employee
So that employee records are stored in the system

Sub User Stories:

1. Validate required employee fields.
2. Auto-generate unique employee ID.
3. Assign department and role.
4. Create login credentials for employee.

User Story 4 – Update Employee Information
As an HR manager
I want to update employee details
So that records remain accurate

Sub User Stories:

1. Edit contact information.
2. Update salary and department.
3. Save modification timestamp.
4. Restrict editing to authorized roles only.

User Story 5 – Deactivate Employee
As an HR manager
I want to deactivate an employee
So that the employee cannot log in but historical data remains

Sub User Stories:

1. Change employee status to “Inactive”.

2. Prevent inactive employee login.
3. Preserve employee data in database.
4. Display inactive status in employee list.

Structured Specification – User Login

Field	**Description**
Function Name	User Authentication
Module	Authentication & Authorization
Description	Authenticates user credentials and grants access to the system
Inputs	Email, Password
Processing	1. Validate input fields are not empty 2. Retrieve user record from database 3. Compare encrypted password 4. Check account status (Active/Inactive) 5. Create session if valid
Outputs	Success: Redirect to dashboard Failure: Display error message
Preconditions	User must be registered in the system
Postconditions	User session created and login time recorded
Exceptions	Invalid credentials, Inactive account, Database connection error

Structured Specification – Add New Employee

Field	Description
Function Name	Create Employee Record
Module	Core Employee Management
Description	Registers a new employee in the system
Inputs	Name, ID, Email, Phone, Department, Role, Salary, Date of Joining
Processing	1. Validate mandatory fields 2. Check email uniqueness 3. Generate unique Employee ID 4. Insert data into Employee table 5. Create login credentials
Outputs	Success confirmation message Validation error message
Preconditions	HR/Admin must be logged in
Postconditions	Employee record and login account created successfully
Exceptions	Duplicate email, Missing required fields

Structured Specification – Update Employee

Field	Description
Function Name	Modify Employee Record
Module	Core Employee Management

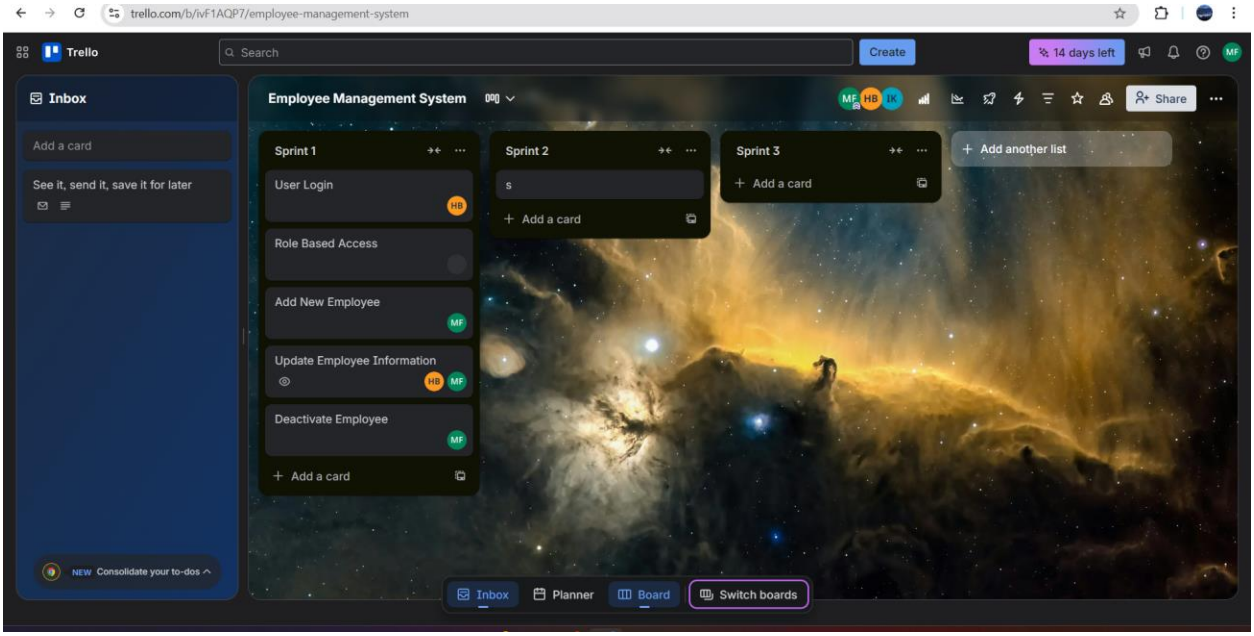
Description	Updates selected employee information
Inputs	Employee ID, Updated fields (Phone, Department, Salary, etc.)
Processing	1. Verify user authorization 2. Retrieve employee record 3. Update specified fields 4. Save modification timestamp
Outputs	Updated record confirmation
Preconditions	Employee must exist; HR/Admin must be logged in
Postconditions	Employee data updated in database
Exceptions	Unauthorized access, Employee not found

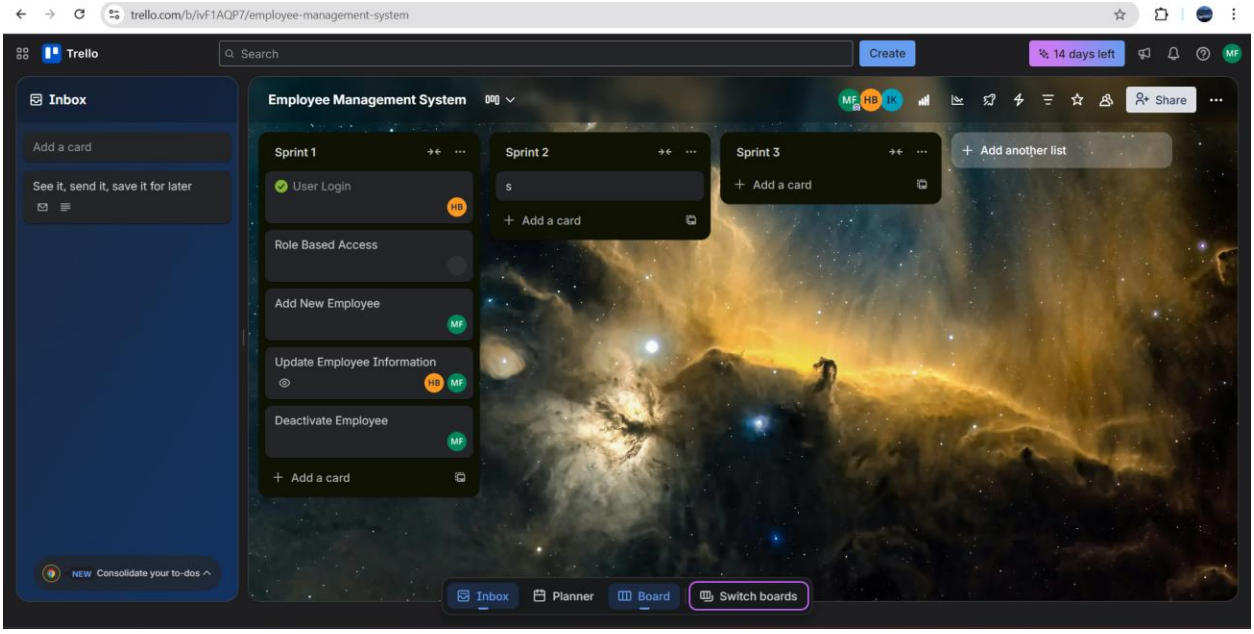
Structured Specification – Deactivate Employee

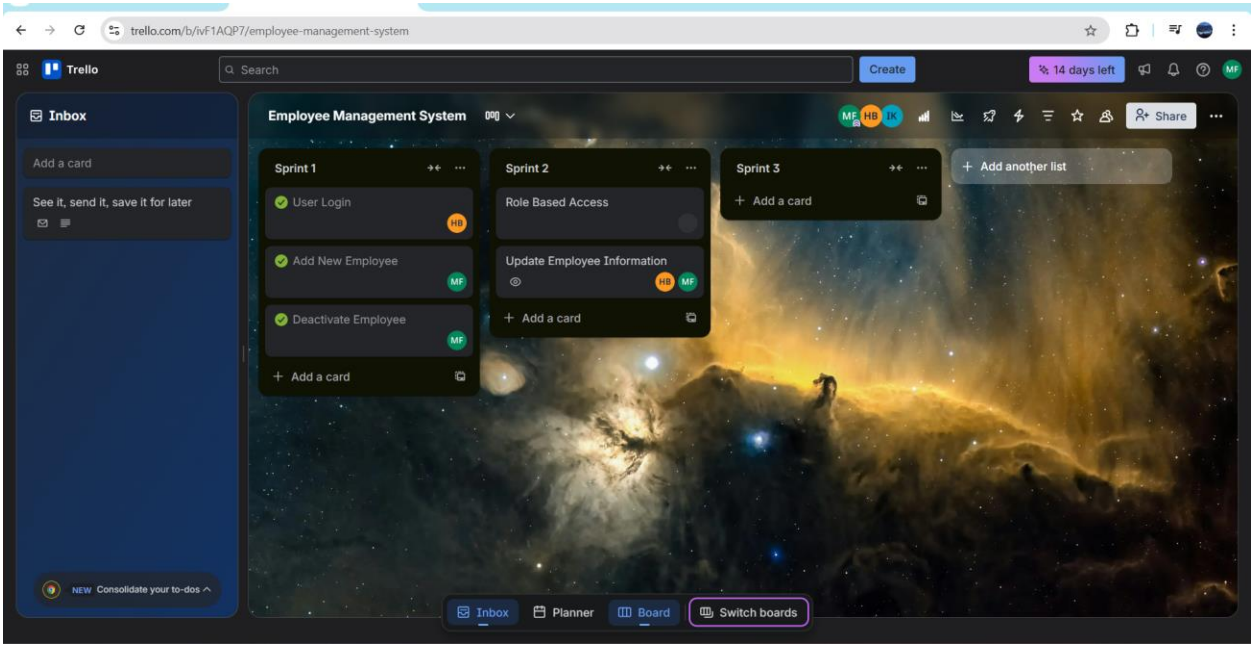
Field	**Description**
Function Name	Deactivate Employee Account
Module	Core Employee Management
Description	Changes employee status to inactive and disables login access
Inputs	Employee ID
Processing	1. Verify authorization 2. Change status to “Inactive” 3. Disable associated login account
Outputs	Confirmation message
Preconditions	Employee must exist

Postconditions	Employee cannot log in; historical records preserved
Exceptions	Employee not found, Unauthorized action

Screenshots:







Complete tasks: User login, Add Employee, Deactivate Employee

Uncompleted: Role-Based Access, Update Employee Info